📘 Used Bike Price Analysis Using Machine Learning

ABSTRACT

This project focuses on performing retail sales analytics on a Supermart Grocery Sales dataset. The objective is to analyze sales trends, category performance, regional contribution, customer purchasing behavior, and profit distribution.

Using Python libraries such as Pandas, Matplotlib, and Seaborn, the dataset was cleaned, processed, and visualized to extract meaningful business insights. The analysis helps in understanding revenue patterns and identifying key contributors to sales and profit.

---

1. INTRODUCTION

Retail businesses generate large amounts of transactional data daily. Analyzing this data helps organizations make informed decisions regarding inventory management, customer targeting, and revenue optimization.

This project aims to apply data analytics techniques to understand sales performance across different categories, regions, and customers.

---

2. DATASET DESCRIPTION

The dataset used is **Supermart Grocery Sales – Retail Analytics Dataset**.

Key Columns:

- Order Date
- Customer Name
- Category
- Region
- Sales
- Profit

The dataset contains transactional records of grocery sales across different regions and customers.

---

3. OBJECTIVES

- To analyze total sales trends over time
- To identify top selling product categories
- To evaluate region-wise sales performance

- To analyze profit distribution
- To identify top contributing customers

---

4. METHODOLOGY

The project was implemented in the following stages:

4.1 Data Loading and Exploration

The dataset was loaded using Pandas library.

df = pd.read_csv("Supermart Grocery Sales - Retail Analytics Dataset.csv")
Initial exploration was performed using:
df.head() df.info() df.isnull().sum()

This step helped understand the structure and quality of the data.

4.2 Data Cleaning

Duplicate records were removed:

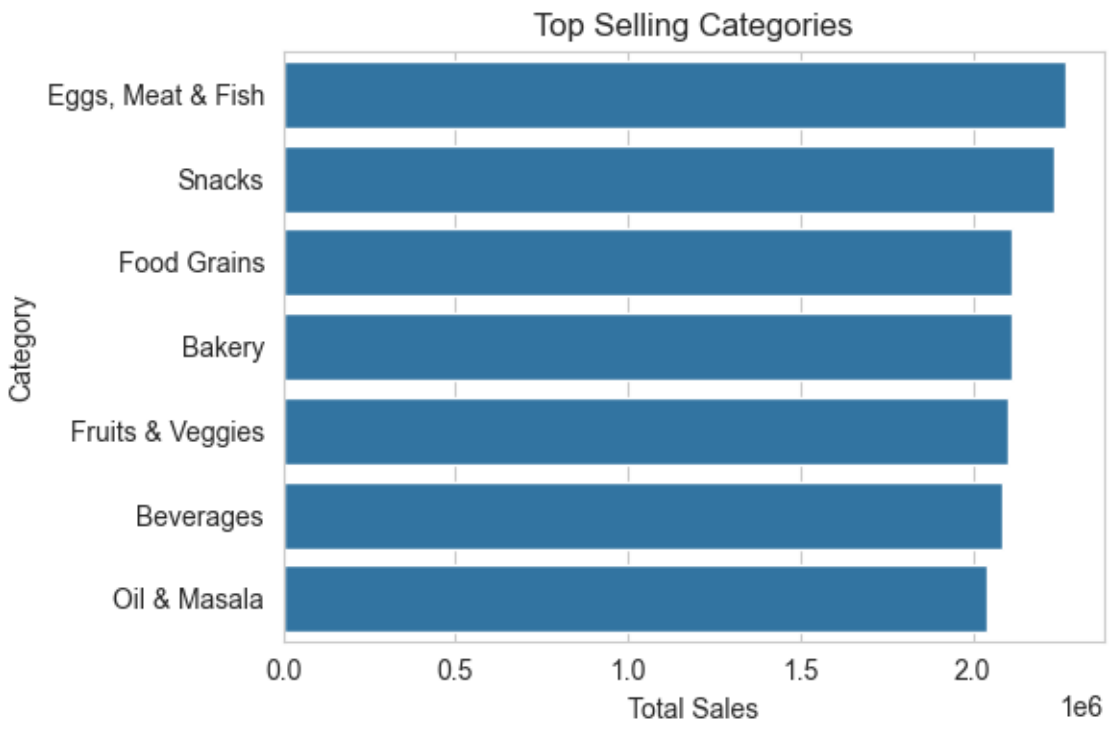
df = df.drop_duplicates()
The Order Date column was converted into datetime format:
df['Order Date'] = pd.to_datetime(df['Order Date'], dayfirst=True, errors='coerce')

This conversion allows time-based analysis.

---

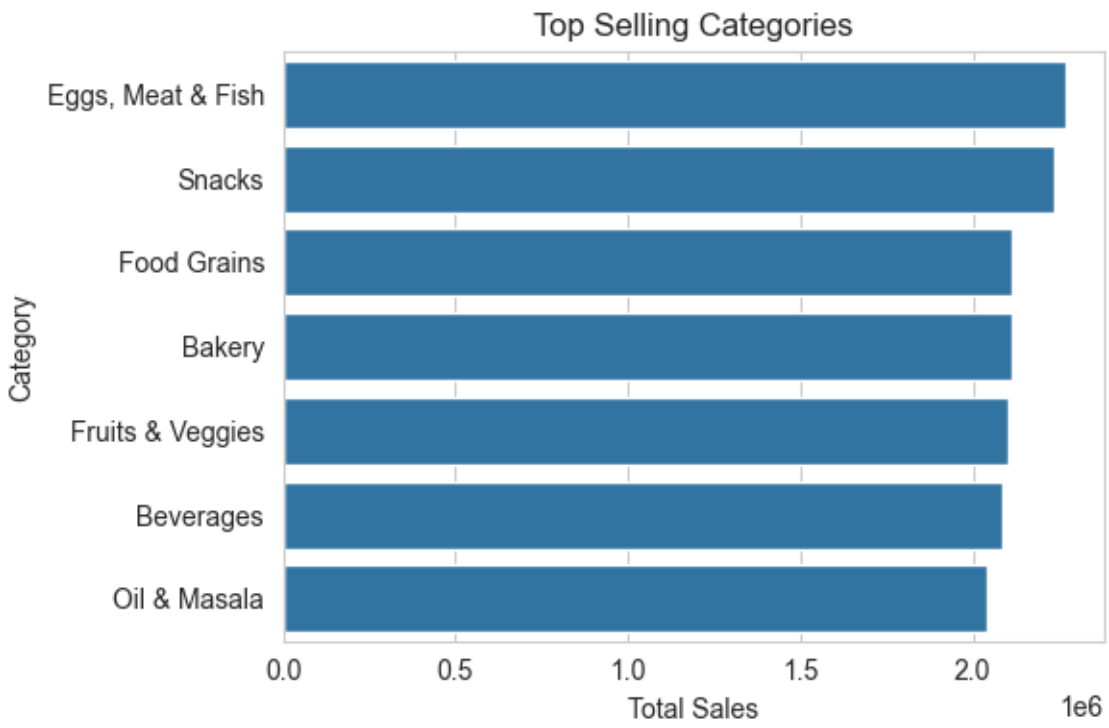
5. DATA ANALYSIS & VISUALIZATION

📊 **Figure 1: Total Sales Over Time**



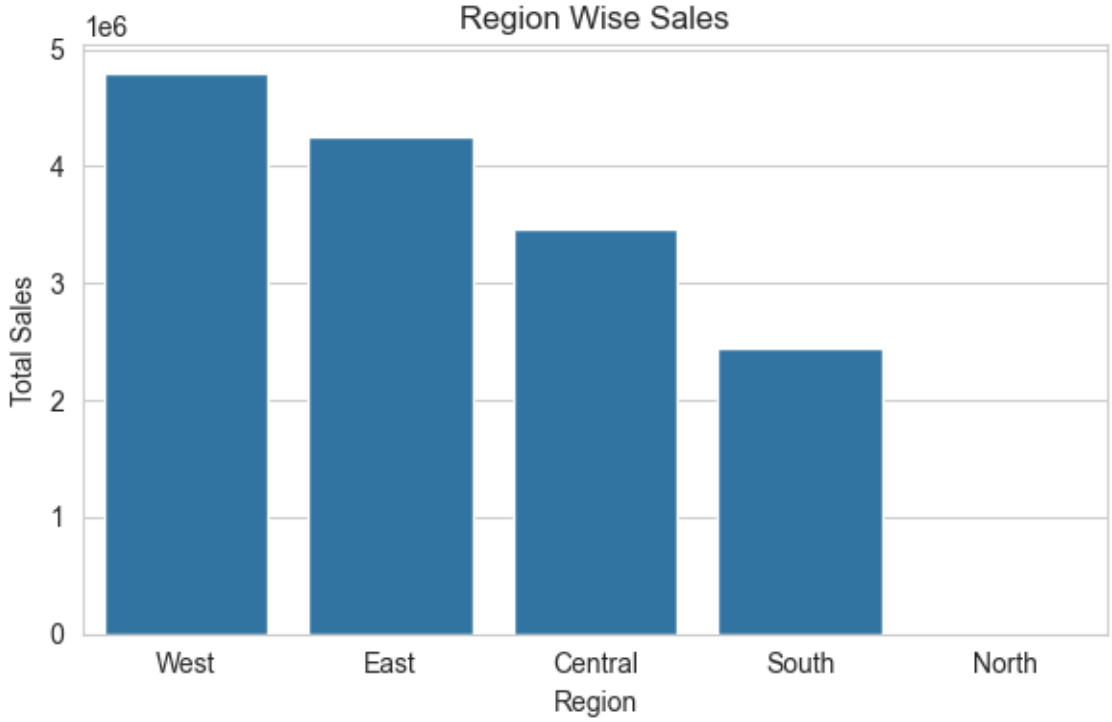
This time-series graph shows how total sales change over time. It helps identify seasonal trends and fluctuations in revenue.

📊 **Figure 2: Top Selling Categories**



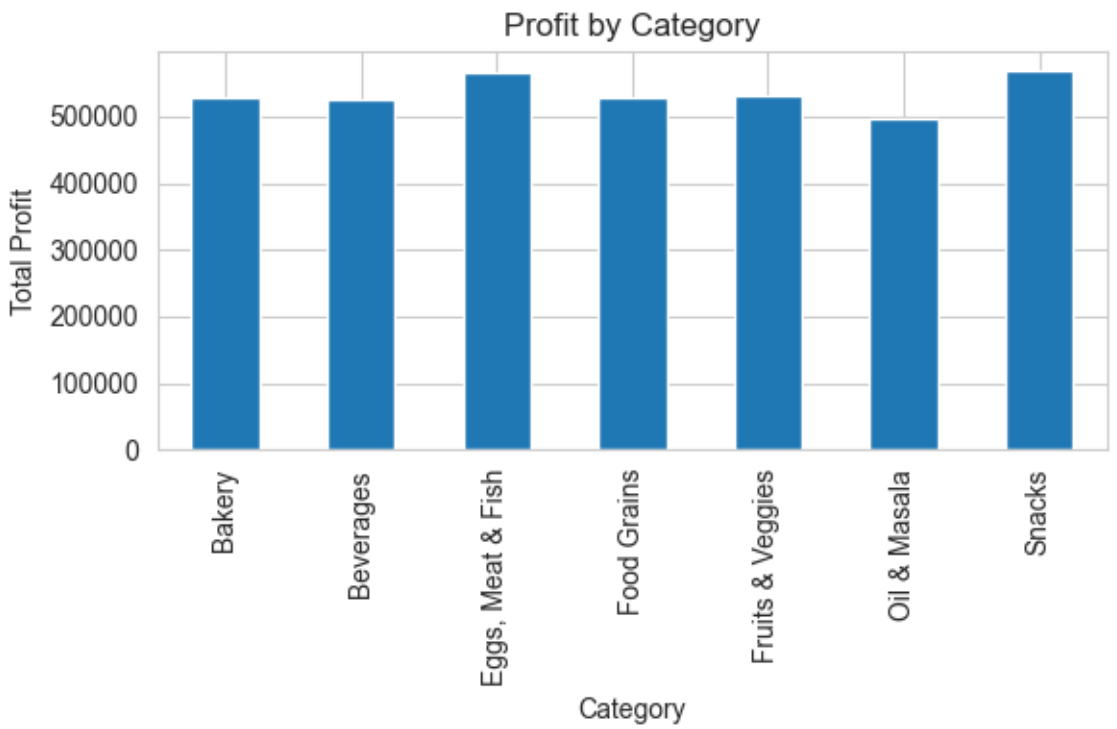
This bar chart represents sales contribution by each product category. It helps identify the most revenue-generating categories.

📊 **Figure 3: Region Wise Sales**



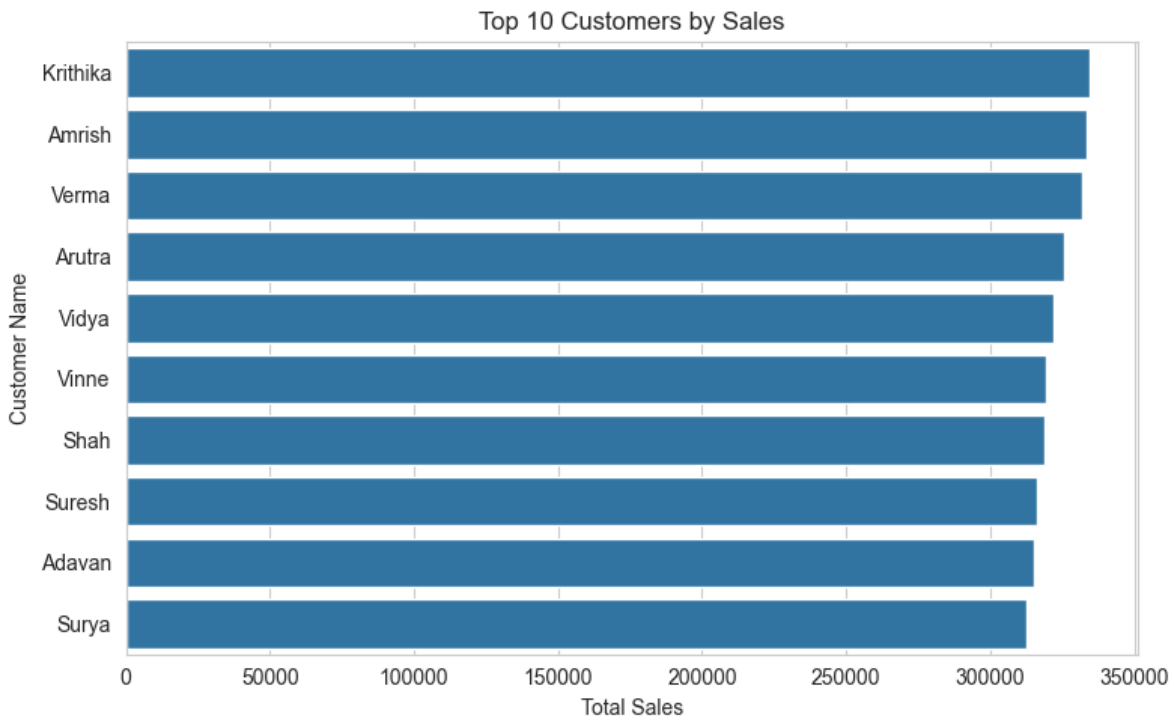
This visualization shows how different regions contribute to overall sales. Some regions clearly generate higher revenue than others.

📊 **Figure 4: Profit by Category**



This graph represents profit distribution across product categories. It helps identify which category is most profitable.

📊 **Figure 5: Top 10 Customers by Sales**



This chart highlights the top 10 customers contributing the highest sales. It indicates that a small number of customers generate a significant portion of revenue.

6. KEY INSIGHTS

- Sales show noticeable variation over time.
- Some categories contribute significantly more revenue.
- Certain regions dominate overall sales performance.
- Profit margins differ across product categories.
- Top 10 customers contribute a major share of revenue.

---

7. BUSINESS APPLICATION

The insights from this analysis can help:

- Improve inventory planning
- Design targeted marketing strategies
- Focus on high-profit categories
- Strengthen operations in high-performing regions
- Build loyalty programs for top customers

---

8. CONCLUSION

The Supermart Grocery Sales analysis demonstrates the power of data analytics in retail business decision-making. Through data cleaning, grouping, aggregation, and visualization, meaningful insights were extracted from raw sales data.

The project highlights how analytical techniques can assist retail companies in improving revenue strategies and understanding customer behavior.

---

9. FUTURE SCOPE

- Build predictive sales forecasting models
- Perform customer segmentation using clustering
- Develop interactive dashboards using Power BI
- Apply machine learning for profit prediction

---

10. TOOLS AND TECHNOLOGIES USED

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn